Why Warm Weather May Have Saved the Crocodiles

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Sixty-six million years ago, an asteroid strike wiped out the dinosaurs. So why did crocodiles survive? New research points to a simple habit: crocodiles and their ancient relatives loved warm weather, and that love may have saved them. Scientists led by University College London, known as UCL, studied more than 7,000 fossils and records from over 400 living and extinct species. They combined that evidence with computer models that rebuilt Earth's climate and geography across 245 million years. Their study was published in Science Advances. The team found a clear pattern. For much of their 250-million-year history, crocodile relatives followed warm climates. When Earth grew hotter, they expanded into new regions, sometimes reaching as far as Antarctica. When temperatures fell, they moved back toward the tropics. That pattern mattered after the asteroid struck. The simulations suggested that tropical and **subtropical** regions stayed suitable for crocodile relatives, even as conditions for many dinosaurs got worse. Areas near the equator could have stayed warm enough to shelter surviving populations, which later recovered and spread out again. Only the most extreme climate simulations removed these **refuges**. Earlier work by members of the same team found that suitable habitats for **non-avian** dinosaurs, the dinosaurs that went extinct, shrank much more severely. Lead author Dr. Alfio Alessandro Chiarenza, based at UCL Earth Sciences, explained it this way: the asteroid did not make the whole planet equally hostile to every group. Crocodiles' long-standing preference for warm places may have put survivors exactly where suitable climates held up best. Their history was also far more varied than today's crocodiles suggest. The wider group, called **pseudosuchians**, included heavily armored plant eaters, land hunters, and giants weighing more than a metric ton, including **Sarcosuchus**, the 12-meter 'super croc.'

Word Treasure

pseudosuchians -- The wider group of crocodile relatives that includes many ancient forms.

subtropical -- Regions near the tropics that stay warm for most of the year.

refuges -- Places where living things can survive when conditions around them get worse.

non-avian -- Describing dinosaurs that are not birds.

Sarcosuchus -- A huge ancient crocodile relative that could grow about 12 meters long.

Background you need

Crocodiles belong to a much wider ancient group called pseudosuchians, which included many different kinds of animals, from heavily armored plant eaters to land hunters and giant predators.

For most of their long history, these animals followed warm climates, spreading into new regions when Earth grew hotter and moving back toward the tropics when it grew cooler.

Warm tropical and subtropical regions could therefore act as refuges where conditions stayed suitable, even when conditions elsewhere became much harder for other animals.

That long-standing preference for warm places may help explain why crocodile relatives survived a major extinction event that wiped out many dinosaurs.

Break it down

LEAD: An asteroid strike wiped out the dinosaurs sixty-six million years ago, yet crocodiles survived.

KEY EVENT: New research published in Science Advances links crocodile survival to a long-standing preference for warm climates.

+- WHO: Scientists led by University College London (UCL), including lead author Dr. Alfio Alessandro Chiarenza, based at UCL Earth Sciences.

+- HOW: They studied more than 7,000 fossils and records from over 400 living and extinct species and combined them with computer models rebuilding Earth's climate and geography across 245 million years.

+- PATTERN: For much of their 250-million-year history, crocodile relatives followed warm climates, expanding when Earth grew hotter, sometimes as far as Antarctica, and moving back toward the tropics when temperatures fell.

+- WHEN: After the asteroid strike, tropical and subtropical regions stayed suitable and could shelter surviving populations, which later recovered and spread out again.

EVIDENCE: Earlier work by members of the same team found suitable habitats for non-avian dinosaurs shrank much more severely, and only the most extreme climate simulations removed the crocodile refuges.

CONTEXT: The wider group, called pseudosuchians, included heavily armored plant eaters, land hunters, and giants weighing more than a metric ton, including Sarcosuchus, the 12-meter super croc.

Why it matters

It shows that surviving a mass extinction can depend on being in places where suitable climates hold up, not only on surviving the disaster itself, and it helps explain why some groups recovered and spread again while non-avian dinosaurs did not.

Test what you remember

1. What wiped out the dinosaurs sixty-six million years ago?

- (A) An asteroid strike
- (B) The loss of all warm climates
- (C) A flood covering Antarctica
- (D) A sudden freeze of the tropics

2. Which institution led the scientists behind the new research?

- (A) A team studying only living crocodiles
- (B) An Antarctic research station
- (C) The journal Science Advances
- (D) University College London (UCL)

3. Across how many million years did the computer models rebuild Earth's climate and geography?

- (A) 66 million years
- (B) 245 million years
- (C) 250 million years
- (D) The last few thousand years

4. When Earth grew hotter, what did crocodile relatives do?

- (A) They stayed only near the equator
- (B) They expanded into new regions, sometimes reaching as far as Antarctica
- (C) They moved back toward the tropics
- (D) They stopped following warm climates

5. What did the simulations suggest about tropical and subtropical regions after the asteroid strike?

- (A) They sheltered only dinosaurs
- (B) They stayed suitable for crocodile relatives
- (C) They were removed in every simulation
- (D) They became hostile to every group equally

6. What did earlier work by members of the same team find about non-avian dinosaurs?

- (A) Their suitable habitats shrank much more severely
- (B) They survived in tropical refuges
- (C) They were unaffected by changing climates
- (D) Their suitable habitats expanded after the asteroid

Answer key (try the quiz first!): 1: A 2: D 3: B 4: B 5: B 6: A

Questions to think about

1. Lead author Dr. Alfio Alessandro Chiarenza: The asteroid did not make the whole planet equally hostile to every group, and crocodiles' preference for warm places may have put survivors exactly where suitable climates held up best.

2. The UCL-led research team: Tropical and subtropical regions stayed suitable for crocodile relatives after the asteroid strike, even as conditions for many dinosaurs got worse.

MY THOUGHTS (at least 20 words):